AGA KHAN UNIVERSITY EXAMINATION BOARD
SECONDARY SCHOOL CERTIFICATE CLASS IX EXAMINATION

Chemistry Paper II (Theory) — Annual 2023

Time: 1 hour 50 minutes | Total Marks: 25

RUBRIC: Answer ALL questions in Section B and Section C. Questions 5 & 6 each offer TWO choices. Attempt any ONE choice from each.

Q.1 (Total 3 Marks)

Draw the structure of an ion with the following features:
• Charge of −3
• Atomic number = 15
• Nucleon number = 31

Q.2 (Total 3 Marks)

Complete the following blanks with reference to the physical properties of labelled elements (Halogens):
• Cl: Odour = _____
• Br: Physical State at Room Temperature = _____
• I: Colour = _____

Q.3 (Total 3 Marks)

When ammonium chloride (NH_4Cl) is heated in a test tube, it results in the formation of ammonia (NH_3) and hydrogen chloride (HCl) gases. The fumes of the gases can be observed moving along the test tube. (Note: N=14, H=1, Cl=35.5)
a. Name the process that allows the movement of gases along the test tube. (1 Mark)
b. Which gas, ammonia or hydrogen chloride, will travel at a faster speed? Give a suitable reason to support your answer. (2 Marks)

Q.4 (Total 4 Marks)

When a piece of Mg ribbon is lit, it burns with a dazzling white light. The combination of Mg with oxygen produces a white substance.
a. Identify the element that has the symbol Mg. (1 Mark)
b. Write a balanced chemical equation to show the given chemical reaction. (2 Marks)
c. Which type of bond is formed in the product of this reaction? (1 Mark)

Q.5 EITHER (Total 6 Marks)

a. An experiment shows that a compound Y contains 80% carbon and 20% hydrogen by mass. If the relative molecular mass of this compound is 30.07 g/mol, then:
i. calculate the empirical formula of the compound Y. (3 Marks)
ii. determine its molecular formula. (3 Marks)
(Note: Atomic mass of C = 12 amu and H = 1 amu)

OR

b. You have a solution of copper(II) sulphate in water. You want to obtain solid copper(II) sulphate from it.
i. Describe why the technique of filtering copper(II) sulphate solution will not work. (1 Mark)
ii. Which method will you use instead? Give a reason to support your answer. (2 Marks)

iii. Describe how you will obtain solid copper(II) sulphate in THREE steps using the method identified in part ii. (3 Marks)

Q.6 EITHER (Total 6 Marks)

a. Consider the given dot and cross structure of a compound: [:Cl:]- (2,8,8) [Ca]2+ (2,8,8) [:Cl:]- (2,8,8).
With reference to the given dot and cross structure:
i. identify the type of bond that exists in it. (1 Mark)
ii. describe the role of each atom in the formation of the bond identified in part i. (2 Marks)
iii. write any THREE general characteristics of such type of compounds. (3 Marks)

OR

b. i. Define the term, ‘electrolysis’. (1 Mark)
ii. Describe the electrolysis of concentrated sodium chloride solution (brine). Support your answer by writing chemical equations for reactions occurring at the cathode and the anode of the electrolytic cell. (5 Marks)